★TOPIC:- HARDWARE

Module_1-Day_5:

❖ Tasks:

1. **List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.**

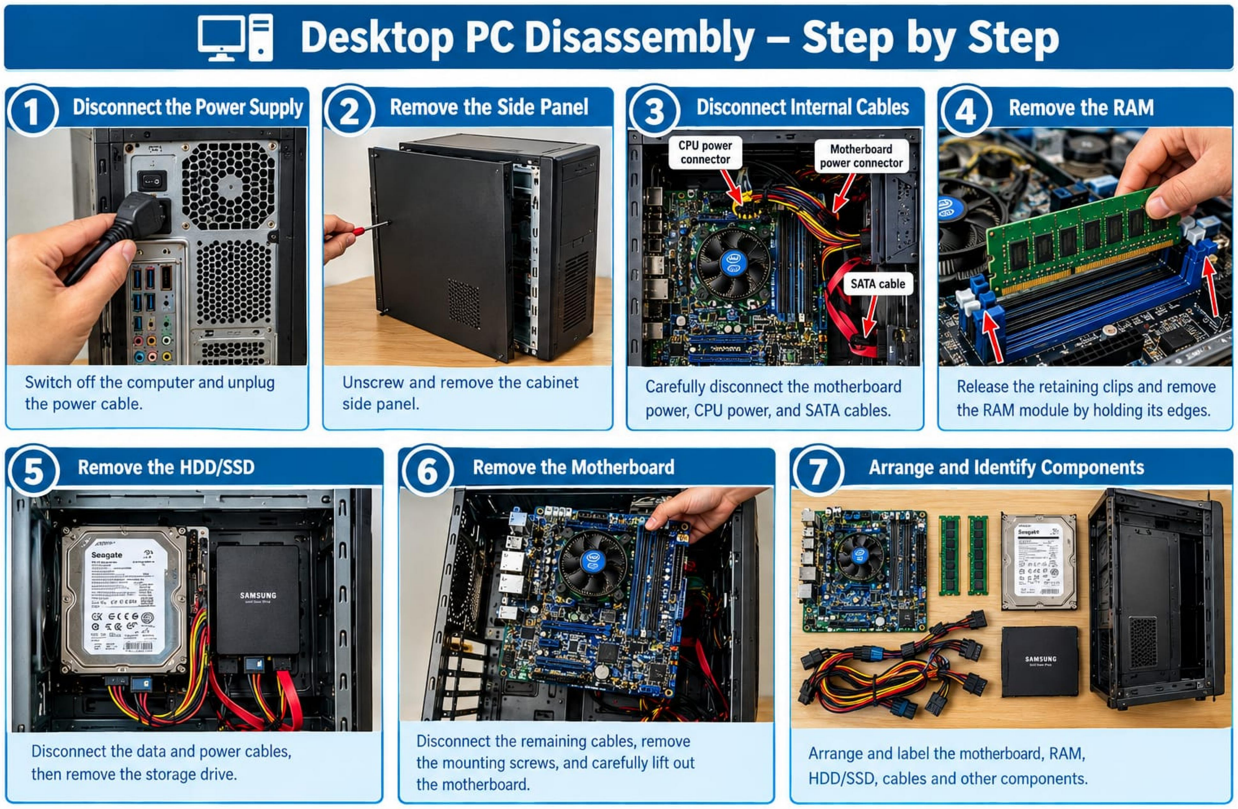
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Component	Description
Motherboard	The main circuit board that connects all components, such as the CPU, RAM, storage, and GPU.
CPU (Central Processing Unit)	The brain of the computer that processes instructions and performs calculations.
RAM (Random Access Memory)	Temporary memory that stores data for currently running programs. It loses its contents when the power is turned off.
HDD (Hard Disk Drive)	A storage device that uses magnetic disks to store files, applications, and the operating system.
SSD (Solid-State Drive)	A fast storage device that uses flash memory to store data and load applications quickly.
GPU (Graphics Processing Unit)	Processes graphics, images, and videos. It may be integrated into the CPU or installed as a dedicated graphics card.
PSU (Power Supply Unit)	Converts electricity from the wall outlet into the required power for the computer's components.
Cooling Fan	Circulates air inside the cabinet to prevent components from overheating.
CPU Heatsink	Absorbs and dissipates heat generated by the CPU, helping maintain a safe operating temperature.
CPU Cooler	Uses a fan or liquid-cooling system to remove heat from the CPU.
CMOS Battery	A small battery on the motherboard that helps maintains the real-time clock and BIOS/UEFI settings when the PC is unplugged.
Expansion Cards	Additional cards, such as sound, network, or graphics cards, that provides extra functionality.
Network Interface Card (NIC)	Allows the computer to connect to a network through Ethernet or, in some cases, Wi-Fi.
SATA and M.2 Connectors	Connect storage devices such as SATA SSDs, HDDs, and M.2 SSDs to the motherboard.
Optical Drive	Reads or writes CDs and DVDs, if an optical drive is installed.
Cabinet (PC Case)	Houses and protects the internal components while providing airflow and mounting points.
Power and Data Cables	Supply electricity to components and transfer data between storage devices and the motherboard.

2. **Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.**

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❖ Desktop PC Disassembly – Step-by-Step:

(1) **Disconnect the Power Supply:** Switch off the computer and unplug the power cable.
(2) **Remove the Side Panel:** Unscrew and remove the cabinet side panel.
(3) **Disconnect Internal Cables:** Carefully disconnect the motherboard power, CPU power, and SATA cables.
(4) **Remove the RAM:** Release the retaining clips and remove the RAM module by holding its edges.
(5) **Remove the HDD/SSD:** Disconnect the data and power cables, then remove the storage drive.
(6) **Remove the Motherboard:** Disconnect the remaining cables, remove the mounting screws, and carefully lift out the motherboard.
(7) **Arrange and Identify Components:** Arrange and label the motherboard, RAM, HDD/SSD, cables, and other components.

3. **Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.**

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❖ Steps to Reassemble a PC:

1. **Install the Motherboard:** Place the motherboard inside the cabinet and secure it with screws. It is installed first because other components connect to it.
2. **Install the CPU:** Carefully place the processor into the CPU socket and lock it. Correct installation ensures the computer can process instructions.
3. **Install the CPU Cooler:** Attach the heatsink and fan, or liquid cooler, to prevent the CPU from overheating.
4. **Install RAM:** Insert the RAM modules into the correct slots until the clips lock. RAM is easier to install before the motherboard area becomes crowded.
5. **Install the Storage Drive:** Connect the HDD or SSD using SATA cables, or install an M.2 NVMe SSD in its slot. Storage holds the operating system and files.
6. **Install the Power Supply (PSU):** Secure the PSU inside the cabinet. It supplies power to all computer components.
7. **Install the Graphics Card (GPU):** Insert the GPU into the appropriate PCIe slot and secure it with screws. Connect its power cable if required.
8. **Connect All Cables:** Connect the motherboard power cable, CPU power cable, storage cables, and cabinet front-panel connectors. Proper connections ensure that every component works correctly.
9. **Install Cabinet Fans and Close the Case:** Check that the fans are connected and cables do not block airflow. Then close and secure the cabinet.
10. **Power On and Test:** Connect the monitor, keyboard, and mouse, then switch on the PC. Check that it boots successfully and recognizes the RAM, storage, and other components.

❖ Why does the order matter?

- Following the correct order makes installation easier, prevents damage to delicate components, avoids blocked connections, and ensures proper cooling and power supply.

4. **Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.**

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❖ Safety Precautions:

Safety precaution	Why it is important
Shut down the PC completely.	Prevents electrical damage and protects components.
Unplug the power cable and all connected devices.	Reduces the risk of electric shock and short circuits.
Disconnect the laptop's internal battery, if applicable.	Prevents accidental power flow while working.
Wear an anti-static wrist strap or touch grounded metal.	Prevents electrostatic discharge (ESD) from damaging components.
Work on a clean, dry, non-carpeted surface.	Reduces static electricity and keeps components safe.
Handle components by their edges.	Prevents damage to delicate pins, contacts, and circuits.
Keep liquids, food, and metal objects away.	Prevents spills, short circuits, and accidental damage.
Never force a component into a slot.	Prevents damage to connectors, sockets, and the motherboard.
Keep screws organized in a small container.	Prevents lost screws and accidental short circuits.
Consult the device's service manual.	Ensures correct procedures for removing and installing parts.

❖ Tools Required:

Tool	Why it is important
Phillips-head screwdriver	Used to remove and tighten cabinet and component screws.
Anti-static wrist strap	Protects sensitive electronic components from static discharge.
Small screw container	Keeps screws and other small parts organized.
Plastic spudger or opening tool	Helps open panels and release clips without scratching or damaging them.
Cable ties or Velcro straps	Organize cables and improve airflow inside the cabinet.
Flashlight	Helps locate small connectors, screws, and slots.
Soft, lint-free cloth	Removes dust without scratching surfaces.
Compressed air or an electronics-safe air blower	Removes dust from fans, heatsinks, and other components.
Thermal paste (if needed)	Helps transfer heat between the CPU and its cooler when the cooler is reinstalled.

❖ Final Checklist before Starting:

- ★ PC is switched off and unplugged.
- ★ All required tools are available.
- ★ Anti-static precautions are in place.
- ★ The workspace is clean, dry, and well-lit.
- ★ Important cables and component connections are documented with photos.
- ★ The service manual is available for reference.